



PasswordStore Audit Report

Version 1.0

0xAdeshina

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- Likelihood & Impact:
- Gas

Protocol Summary

A smart contract application for storing a password. Users should be able to store a password and then retrieve it later. Others should not be able to access the password.

Disclaimer

The 0xAdeshina team makes all effort to find as many vulnerabilities in the code in the given time period, but holds no responsibilities for the findings provided in this document. A security audit by the team is not an endorsement of the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the Solidity implementation of the contracts.

Risk Classification

		Impact		
		High	Medium	Low
Likelihood	High	H	H/M	M
	Medium	H/M	M	M/L
	Low	M	M/L	L

We use the CodeHawks severity matrix to determine severity. See the documentation for more details.

Audit Details

commit hash goes here

Scope

```
1 ./src/  
2 #-- PasswordStore.sol
```

Roles

- Owner: The user who can set the password and read the password.
- Outsiders: No one else should be able to set or read the password

Executive Summary

*Audit summuray goes here, types of things found. *We spent X hours with Z auditors using Y tools.

*this goes well for a team, i believe this will be good.

Issues found

*this is where we put our severities and numbers of issues found

Severity	Number of issues found
High	2
Medium	0
Low	0
Info	1
Total	3

Likelihood & Impact

- Impact: HIGH
- Likelihood: HIGH
- Severity: HIGH

[H-2] PasswordStore::setPassword has no access control. Anyone can call the function and change owners password

Description:

The `PasswordStore::getPassword` function allows for non-owner to change the password, the function is set to be an `external` function. Owners password can be changed by anyone by simply just calling the function with parameters. The natspec of the function states that `This function allows only the owner to set a new password.`

```
1
2     function setPassword(string memory newPassword) external {
3 >> // @audit - There are no access controls
4     s_password = newPassword;
5     emit SetNetPassword();
6     }
```

seen as there is no access control against non-owner calls, the smart contract application is vulnerable to user's password being changed

Impact: Anyone can set/change the password of the contract, severely breaking the contract's intended functionality

Proof of Concept: Add the following to the `PasswordStore.t.sol` test file.

Code

```
1
2 function test_anyone_can_set_password(address randomAddress) public {
3 vm.assume(randomAddress != owner);
4 vm.prank(randomAddress);
5 string memory expectedPassword = "myNewPassWord";
6 passwordStore.setPassword(expectedPassword);
7
8     vm.prank(owner);
9     string memory actualPassword = passwordStore.getPassword();
10
11     assertEq(expectedPassword, actualPassword);
12 }
```

Recommended Mitigation:

```
1 if(msg.senderr != s_owner)
2     revert PasswordStore_notOwner();
```

Likelihood & Impact:

- Impact: HIGH
- Likelihood: HIGH
- Severity: HIGH

Medium**Low****Informational**

[S-#] The PasswordStore::getPassword natspec indicates a parameter that doesn't exist, causing the natspec to be incorrect

Description:

```
1 ```javascript
2
3 /*
4  * @notice This allows only the owner to retrieve the password.
5  * @param newPassword The new password to set.
6  */
7 function getPassword() external view returns (string memory)
8
9 ```
```

The `PasswordStore::getPassword` function signature is `getPassword()` which the natspec say it should be `getPassword(string)`

Impact: The natspec is incorrect

Proof of Concept:

Recommended Mitigation: Remove the incorrect natspec line.

```
1
2 -      * @param newPassword The new password to set.
```

Likelihood & Impact:

- Impact: NONE
- Likelihood:
- Severity: Information/Gas/Non-crits

Gas